

Problem 5

On the first third of the total distance, a car moves at a speed of v_1 , while on the remaining two-thirds of the distance, it moves at a speed of $v_2 = 54 \text{ km/h}$. The average speed of the car over the entire journey is $\langle v \rangle = 36 \text{ km/h}$. What is the speed v_1 ?

Solution

The average speed is given by:

$$\langle v \rangle = \frac{\text{Total distance}}{\text{Total time}}.$$

The total distance s is divided as:

$$s_1 = \frac{s}{3}, \quad s_2 = \frac{2s}{3}.$$

The time spent in each segment is:

$$t_1 = \frac{s_1}{v_1} = \frac{\frac{s}{3}}{v_1} = \frac{s}{3v_1},$$

$$t_2 = \frac{s_2}{v_2} = \frac{\frac{2s}{3}}{54} = \frac{s}{81}.$$

The total time is:

$$t_{\text{total}} = t_1 + t_2 = \frac{s}{3v_1} + \frac{s}{81}.$$

The average speed is:

$$\langle v \rangle = \frac{s}{t_{\text{total}}} = \frac{1}{\frac{1}{3v_1} + \frac{1}{81}}.$$

Substitute $\langle v \rangle = 36 \text{ km/h}$ and $v_2 = 54 \text{ km/h}$:

$$36 = \frac{1}{\frac{1}{3v_1} + \frac{1}{81}}.$$

Simplify:

$$\frac{1}{36} = \frac{1}{3v_1} + \frac{1}{81}.$$

Find a common denominator:

$$\frac{1}{36} = \frac{27}{81v_1} + \frac{1}{81}.$$

Multiply through by $81v_1$:

$$\frac{81v_1}{36} = 27 + v_1.$$

Simplify:

$$2.25v_1 = 27 + v_1.$$

Rearrange:

$$1.25v_1 = 27.$$

Solve for v_1 :

$$v_1 = \frac{27}{1.25} = 21.6 \text{ km/h}.$$

Conclusion

The speed v_1 is:

$$v_1 = 21.6 \text{ km/h}.$$